

Configuration documentation

Configuration documentation

System settings, versions, and how to change them without losing the ability to roll back.

Runtime configuration

Setting	Default	How to change
HTTP port (local)	3847	Environment variable <code>PORT</code>
Local data folder	<code>backend/var/</code>	Environment variable <code>DATA_DIR</code>
Session cookie	<code>ccst_session</code>	<code>backend/middleware/auth.js</code>
Default instructor	<code>instructor / ProCloud-G18</code>	<code>backend/data/seed-data.js</code> , then <code>npm run seed</code>

Student accounts are not configuration. An instructor signs up, then adds students under **Class & students** with a password they choose. Do not edit hash strings in the JSON store by hand.

Environment variables (deployed)

Production runs on **Azure App Service** (`ccst-ticketing`, Central India). Set these under **Configuration** → **Application settings** in the Azure portal, or via `npm run deploy:azure / az webapp config appsettings set`.

Variable	Required	What it does
<code>DATABASE_URL</code>	Preferred in production	Postgres connection string. When set, the app stores state in Azure Database for PostgreSQL (<code>documents</code> JSONB table) instead of the blob.
<code>PERSIST_BACKEND</code>	Optional	Force <code>sql</code> or <code>json</code> . Default: use SQL when <code>DATABASE_URL</code> is set, otherwise Azure blob / file. Set <code>json</code> to roll back to blob without removing <code>DATABASE_URL</code> .
<code>AZURE_BLOB_SAS_URL</code>	Yes if not on SQL	Container SAS URL for the blob holding <code>db.json</code> . Keep it even on SQL for backups / rollback. Without SQL or blob, the app falls back to memory and every save is silently lost.
<code>AZURE_STATIC_BASE_URL</code>	Yes in production	Public base for docs PDFs/images, e.g. <code>https://ccstticketing.blob.core.windows.net/docsassets</code> . Upload with <code>npm run upload:static</code> .
<code>SESSION_SECRET</code>	Yes on App Service	Signs session cookies; generate a long random string and keep it stable across redeployments. Production refuses to start if this is missing or still the built-in default.
<code>NODE_ENV</code>	Recommended	Set to <code>production</code> on App Service
<code>INSTRUCTOR_SIGNUP_CODE</code>	Yes, to allow instructor signup	Shared secret instructors must enter on Instructor signup . Without it, signup stays closed. Change it anytime in App Service settings; do not publish it in student docs.
<code>BOOTSTRAP_INSTRUCTORS</code>	Optional	JSON array of instructor accounts to create on first read, so a fresh deployment is not locked out
<code>BLOB_READ_WRITE_TOKEN</code>	Optional	Legacy Vercel Blob storage, used only if neither SQL nor <code>AZURE_BLOB_SAS_URL</code> is set

Static docs (PDFs and pictures)

Classroom Documentation PDFs and thumbnail art are stored in Azure Blob container `docsassets` (public read), not on the App Service disk.

```
npm run upload:static
```

That creates/updates `docs-pdf/` and `docs-art/` under the container and prints `AZURE_STATIC_BASE_URL`. Deploy sets the same setting on the web app.

Source files live in `backend/docs-assets/` (`docs-md`, `docs-pdf`, `docs-art`) for rebuild/upload only. The running app loads docs from Azure Blob; it does not need that folder.

Pre-class checks

- Follow `backend/docs-assets/docs-md/13-instructor-class-day.md` (class-day sheet).
- Open `https://ccst.website/api/health` — expect `"ok": true`, `blob0k`, `sessionSecret0k`, `static0k`, and `classroomAiReady: true`.
- Snapshot blob state: `npm run backup:azure` (copies `db.json` and `lab/*.json` under `state/backups/<timestamp>/`).
- Day-to-day ops (health, deploy, backups, rollback): `backend/docs-assets/docs-md/16-ops.md` (trainer notes — not listed under in-app Documentation).
- Export marks from the app anytime: **Class & students** → **Export CSV**.

- Default G18 login: `instructor` / `ProCloud-G18` . Hard-refresh once so clients load current JS modules.

Deployment slots

The current plan is **B1**. Azure deployment slots need **Standard (S1)+** (extra cost). Stay on B1 and deploy outside class hours with `npm run deploy:azure` .

Instructor accounts and passwords

The seed login `instructor` / `ProCloud-G18` is a **lab convenience**. Change it for any shared or long-lived deployment, and rotate `INSTRUCTOR_SIGNUP_CODE` if it was ever shared. Do not publish the signup code in student handouts.

Keep the App Service and the storage account in the **same Azure region** (Central India) — see `02-infrastructure.md` .

For **Vercel** deployments, the function region is pinned in `vercel.json` (`"regions": ["bom1"]`) and should stay near the blob.

Storage picks itself in this order: Azure blob, then Vercel Blob, then a local file, then memory. Only the last one loses data, and the app puts an orange banner on every page when it lands there.

Device and cabling state

The Lab map is configuration too, and it is per class:

What	Reset with	Note
Cabling	Reset to design on the Lab map	Keeps cables unplugged for lab-map tickets that are still open
Device configuration	Same button	Hostnames, addresses, VLANs, routes and NAT go back to the shipped design

Shipped cabling and configuration live in `server/domain/lab/lab-map.js` and `server/domain/lab/net-sim.js` . Both are versioned (`MAP_SCHEMA` , `STATE_SCHEMA`); bumping a version migrates existing classes instead of wiping the work they have done.

Software versions (lab)

- Node.js 18 or newer.
- Express 4.x.
- cookie-parser 1.x.

There is no separate application version table in a database. The app version in `package.json` is `1.0.0` .

Ticket numbering

Tickets are `TKT-0001` , `TKT-0002` , ... from `nextTicket` in the JSON store. After a seed, numbering starts again at 1.

SLA numbers (change with care)

Stored in each class as `slaPolicy` (defaults from `server/data/seed-data.js` on first create). Instructors edit response and resolve times under **Class & students** → **Class SLA**.

Priority	Response	Resolve
Critical	30 minutes	6 hours
High	1 hour	12 hours
Medium	2 hours	24 hours
Low	3 hours	48 hours

If you change the policy, re-seed or the dashboard will still use the old copy inside `db.json`.

Rollback

- **Code:** use git history if the project is under version control. On Azure App Service, redeploy a previous zip or use deployment slots. On Vercel, promote an earlier deployment.
- **Data (deployed):** download the blob before experiments and upload it again to restore. Sessions reset either way.
- **Data (local):** keep a copy of `data/db.json`, replace the file, restart.
- **Bad seed:** delete `data/db.json` and start the server; a fresh one is created automatically.

Dependencies of a typical ticket

A ticket points at:

- `requesterId` → `requesters[]`
- `assigneeId` → `users[]`
- `comments[].authorId` → `users[]`

Do not delete users from JSON while tickets still reference them. Re-seed instead.